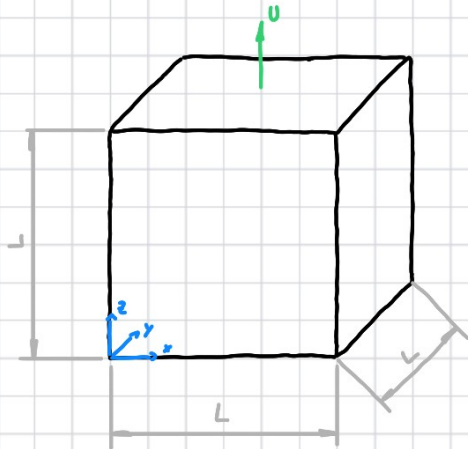


Odgen Material Model



8452 silicone rubber

$$\dot{\epsilon} = 0,4 \text{ } 1/s$$

$$\alpha = 3$$

$$\mu = 0,4$$

$$L = 10 \text{ mm}$$

Odgen

$$\Psi = \frac{2\mu}{\alpha^2} (\lambda_1^\alpha + \lambda_2^\alpha + \lambda_3^\alpha - 3)$$

$$\underline{\underline{C}} = \underline{\underline{F}}^T \underline{\underline{F}}$$

↓ Eigen values

$$c_i = \lambda_i^2$$

invariant based Odgen

$$J = \det(\underline{\underline{F}})$$

$$\underline{\underline{C}} = \underline{\underline{F}}^T \underline{\underline{F}}$$

$$\bar{J}_1 = J^{-2/3} \text{tr}(\underline{\underline{C}})$$

$$\Psi = \Psi_{iso} + \Psi_{vol}$$

$$\Psi_{iso} = \frac{\mu}{2\beta 3^{\beta-1}} (\bar{J}_1^\beta - 3^\beta)$$

$$\beta = \frac{\alpha}{2}$$

$$\Psi_{vol} = \frac{1}{2} K (J - 1)^2$$

$$K = \frac{2\mu(1+\nu)}{3(1-2\nu)} \quad \nu_{pen} = 0,495$$

Stress and stretch

$$\underline{\underline{P}} = \frac{\partial \Psi}{\partial \underline{\underline{F}}}$$

$$R_z = \int_{\Gamma_{top}} (\underline{\underline{P}} \cdot \underline{\underline{N}}) \cdot \underline{\underline{e}}_z dA$$

$$\sigma = \frac{R_z}{A_0} \quad A_0 = L^2$$

$$\epsilon_{eng} = \frac{u_z}{L}$$

$$\lambda = 1 + \epsilon_{eng}$$